Simplifying Algebraic Fractions

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Simplifying Algebraic Fractions

What is an algebraic fraction?

- A fraction with a numerator or denominator (or both) that contains an **algebraic term** is an **algebraic fraction**
- For example
  - $\frac{x}{4}$ or $\frac{48}{y^2}$ or $\frac{2p^3}{4p^2+1}$
- You need to be able to **simplify** basic algebraic fractions
  - Like $\frac{x^5}{x^2}$ or $\frac{3}{6p}$

How do I simplify algebraic fractions?

- To simplify any **powers** (indices) it helps to **write out the multiplications in full**
  - $\frac{n^5}{n^2} = \frac{n \times n \times n \times n \times n}{n \times n}$
- Terms can then be "**cancelled**"
  - $\frac{n^5}{n^2} = \frac{n \times n \times n \times \cancel{n} \times \cancel{n}}{\cancel{n} \times \cancel{n}} = n \times n \times n = n^3$
  - $\frac{m^2}{m^3} = \frac{m \times m}{m \times m \times m} = \frac{\cancel{m} \times \cancel{m}}{m \times \cancel{m} \times \cancel{m}} = \frac{1}{m}$
- Any **numerical** fractions can be simplified as usual
  - $\frac{4}{12} = \frac{2}{6} = \frac{1}{3}$
- You also need to be familiar with **alternative ways of writing fractions**
  - $\frac{3x}{4}$ is the same as $3 \times \frac{x}{4}$ or $\frac{3}{4}x$ or $0.75x$
  - $\frac{5}{6y}$ is the same as $5 \times \frac{1}{6y}$ or $\frac{5}{6} \times \frac{1}{y}$

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

(a) $\frac{4p^3}{12p^6}$

Separate the numerical part from the algebraic part

$$\frac{4}{12} \times \frac{p^3}{p^6}$$

Simplify the numerical fraction

$$\frac{1}{3} \times \frac{p^3}{p^6}$$

Write out the algebraic terms in full

$$\frac{1}{3} \times \frac{p \times p \times p}{p \times p \times p \times p \times p \times p}$$

Cancel any terms that appear in both the numerator and denominator

$$\frac{1}{3} \times \frac{\cancel{p} \times \cancel{p} \times \cancel{p}}{p \times p \times p \times \cancel{p} \times \cancel{p} \times \cancel{p}}$$

$$\frac{1}{3} \times \frac{1}{p \times p \times p}$$

Simplify

$$\frac{1}{3} \times \frac{1}{p^3}$$

Combine into one term

$$\mathbf{\frac{1}{3p^3}}$$

(b) $\frac{32q^5}{8q^2}$

Separate the numerical part from the algebraic part

$$\frac{32}{8} \times \frac{q^5}{q^2}$$

Simplify the numerical fraction

$$4 \times \frac{q^5}{q^2}$$

Write out the algebraic terms in full

$$4 \times \frac{q \times q \times q \times q \times q}{q \times q}$$

Cancel any terms that appear in both the numerator and denominator

$$4 \times \frac{q \times q \times q \times \cancel{q} \times \cancel{q}}{\cancel{q} \times \cancel{q}}$$

$$4 \times q \times q \times q$$

Simplify

$$\mathbf{4q^3}$$

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